Project Abstract

Title : Energy Consumption Analytics Dashboard

Commercial buildings combine HVAC, lighting, plug loads, rooftop photovoltaic generation and battery storage. Their operation changes with weather, occupancy and electricity prices. This project presents a **stateful building-energy digital twin** for a 7,500 m² reference office that connects telemetry, thermal modeling, forecasting, fault detection, scenario analysis and comfort-constrained control.

The project uses real external drivers for Berlin in 2019: ERA5 outdoor weather, PVGIS-SARAH2 irradiance and PV output for a 250 kWp system, and German/Luxembourg day-ahead prices from an Open Power System Data snapshot. The joined driver series contains 8,758 hourly UTC records; two hours without a published price are omitted. The building interior—including occupancy, indoor temperature, HVAC, lighting and plug loads—is **simulated; no live building or real meter is connected.**

A telemetry-only twin maintains timestamped state and identifies thermal parameters. Ridge regression is selected for outdoor-temperature and base-load forecasts, while a seasonal-mean model is selected for irradiance; model selection covers 1,032 samples. Residual and flatline detectors are assessed with injected faults. A Pyomo/HiGHS optimizer runs a **24-hour comfort-constrained MPC** loop for HVAC and a 250 kWh battery. A local Streamlit dashboard presents state, forecasts, fault results, scenarios and control comparisons.

In the repository's 78-day comparison, cost-focused MPC reduces operating cost from €5,868 to €5,440 (7.3%) and records zero comfort violations versus eight for the thermostat. HVAC electricity rises from 21.81 to 26.24 MWh

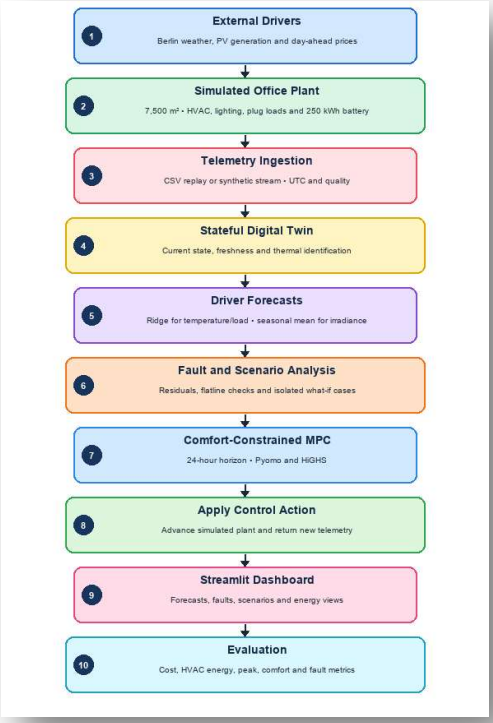
(+20.3%) in the repository results table. The presentation and decision log report +33.2%, so the source figures require reconciliation. Scenario results report €428 baseline savings, €57 with no battery and €720 with doubled battery capacity. A flatline detector reaches 0.944 recall with four-hour latency on the injected stuck-temperature-sensor case. These are **simulation outcomes, not measured savings from a real building.**

Datasets

Dataset	URL link
Hybrid driver series: Berlin 2019 weather/PV (PVGIS/ERA5) and DE/LU day-ahead prices (OPSD).	PVGIS/JRC: re.jrc.ec.europa.eu/pvg_tools/en/ https://archive.ics.uci.edu/dataset/320/student+performance OPSD: data.open-power-system-data.org/time_series/2020-10-06

Approach

Hourly exogenous drivers are aligned in UTC and replayed through a documented single-zone office simulator. Quality-tagged telemetry updates the twin, which estimates thermal behavior without reading simulator internals. The pipeline selects forecasts, checks injected faults, runs isolated what-if cases and solves a 24-hour HVAC/battery schedule subject to comfort constraints. The first MPC action advances the simulated plant and its next telemetry updates the twin. Results are reviewed using operating cost, HVAC energy, peak demand, comfort violations, forecast error, and fault recall/latency.



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